

Analysis First-Year Exam, August 2020, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. The function $f : [0, 1] \rightarrow \mathbb{R}$ is Riemann-integrable and vanishes at each point of a dense set. Does it follow that $\int_0^1 f = 0$? Proof or counterexample, as appropriate.
2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Prove that there exists a sequence (p_n) of polynomials that converges to f both
 - (i) pointwise on $\mathbb{R}$
 - (ii) uniformly on each compact set in $\mathbb{R}$ (simultaneously).
3. Let $\mathcal{F} \subseteq C(X)$ be equicontinuous and let B be the set comprising all points of X at which $\mathcal{F}$ is bounded; that is, $a \in B$ exactly when there exists K such that $|f(a)| < K$ whenever $f \in \mathcal{F}$. Prove that $B \subseteq X$ is both closed and open.
4. Let (f_n) be a sequence of measurable real-valued functions on Ω . Prove that each of the following subsets of Ω is measurable:
 - (i) $P = \{\omega : (f_n(\omega)) \text{ converges to an irrational number}\}$
 - (ii) $Q = \{\omega : (f_n(\omega)) \text{ converges to a rational number}\}$
 - (iii) $R = \{\omega : (f_n(\omega)) \text{ converges to a real number}\}$.
5. Let (f_n) be a sequence of bounded, measurable, real-valued functions on Ω and assume that $f_n \rightarrow f$ uniformly. Prove that if $\mu(\Omega) < \infty$ then

$$\int_{\Omega} f_n d\mu \rightarrow \int_{\Omega} f d\mu$$

and show by example that this can fail if $\mu(\Omega) = \infty$.